

Exercise 1

Defining dynamic treatment strategies

Learning Objectives

By the end of this Exercise, be able to map a pharmacoepidemiologic research question to the protocol of a target trial, and understand clinical implications of design choices.

Overview

Questions will be based on the paper Glucagon-like peptide-1 receptor agonists and risk for cardiovascular events in older adults treated with levothyroxine: a target trial emulation, by Du et al. 2025. Reading this paper is not required for participation in this exercise, but you may choose to do so for additional background context. A brief description has been included for your convenience.

Background: Initiation of Glucagon-Like Peptide-1 receptor agonists (GLP-1RA) in patients with type 2 diabetes (T2D) treated with levothyroxine may decrease the required levothyroxine dose due to weight loss or enhance levothyroxine absorption through delayed gastric emptying. These changes may cause thyroid hormone over-replacement and increased risk of atrial fibrillation/flutter (AF/Aflutter) and stroke. This study aims to investigate the impact of GLP-1RA compared to SGLT2 inhibitors on risks of AF/Aflutter and stroke in patients with T2D treated with levothyroxine.

Methods: We used weighted Cox proportional hazards models to estimate the hazard ratio (HR) and 95% confidence interval (CI) for primary and secondary outcomes. Inverse probability weights (IPW) were estimated by pooled logistic regression to account for informative censoring due to treatment discontinuation, treatment initiation, bariatric surgery, initiation of medications suggestive of thyroid disease or interfere with thyroid hormone levels, new diagnosis of a thyroid neoplasm, thyroid cancer, hyperthyroidism, thyroiditis, or other thyroid diseases.

Results: After matching, the study cohort included 2,384 participants in both GLP-1RA and SGLT2 groups. The median follow-up time was 1.05 years. Compared to SGLT2 inhibitors, initiation of GLP-1RA was associated with higher risk of AF/Aflutter (HR: 1.46;

95% CI: 1.28–1.67), while no statistically significant difference was observed between the two groups for stroke (HR: 1.17; 95% CI: 0.98–1.39).

Part 1

Below find the treatment, outcome, and follow-up aspects from the protocol of the target trial Du et al. 2025 aimed to emulate using Medicare data.

	Specification	Emulation
Eligibility	• Age ≥ 65 • Diagnosed with T2D • Stable levothyroxine dose for ≥ 6 months • Willingness to participate and provide informed consent	• Same as the target trial • The base-cohort entry date was defined as the date of patients' first inpatient diagnosis or the second outpatient diagnosis of T2D, identified by relevant ICD-9/10-CM diagnosis codes • Patients entered the study-cohort at the date when they initiate GLP-1RA or SGLT2 inhibitors
Treatments	• Eligible patients were randomly assigned to initiate GLP-1RA or SGLT2 inhibitors in a 1:1 ratio	• We emulated randomization using PS matching with a 1:1 ratio to initiate GLP-1RA or SGLT2 inhibitors
Follow-up	• Patients were followed from the date of randomization until the first occurrence of any of the following events: a study outcome, bariatric surgery, initiation of medications suggestive of thyroid disease or interfere with thyroid hormone levels (e.g., PPIs, prednisone, or prednisolone), new diagnosis of a thyroid neoplasm, thyroid cancer, hyperthyroidism, thyroiditis, or other thyroid diseases (excluding hypothyroidism), loss to follow-up, discontinuation of assigned treatment (GLP-1 or SGLT2), switch to the other studied treatment, death, or the end of the study period (December 31, 2020)	• Same as the target trial • Loss to follow-up was defined by not continuous enrolling in the Medicare Part A/B/D plan

Based on the study protocol, answer the following questions:

- a) What is the target population? Do the eligibility criteria ensure that all patients could plausibly receive either treatment strategy?

Based on the eligibility criteria, the intended target population includes individuals with type 2 diabetes above the age of 65 on stable dose of levothyroxine.

The eligibility criteria are broad such that some patients may be contraindicated for GLP-1RA or SGLT2, for example, individuals with pancreatitis or medullary thyroid cancer are not recommended to use GLP-1RA agents.

- b) Recall that for certain estimation procedures, like inverse probability weighting, individuals may be censored when they deviate from the assigned strategy. With this in mind, is the study estimating the effect of GLP-1RA vs. SGLT2 inhibitors alone, or is it estimating something more specific?

Follow-up ended not just at treatment changes (i.e., addition of GLP-1 to SGLT2 or discontinuation of SGLT2) but also at bariatric surgery, initiation of medications suggestive of thyroid disease or interfere with thyroid hormone levels, new diagnosis of a thyroid neoplasm, thyroid cancer, hyperthyroidism, thyroiditis, or other thyroid diseases.

Censoring implies a deviation from the treatment strategy, meaning the treatment strategies are more complicated than initially outlined in the table. The investigators are estimating the effect of initiating and continuing GLP1 without bariatric surgery, initiating medications suggestive of thyroid disease, developing thyroid cancer, etc. vs initiating and continuing SGLT2 without bariatric surgery, initiating medications suggestive of thyroid disease, developing thyroid cancer, etc.

Part 2

Consider the following proposed modifications to the study design, which change one or more components of the original protocol. For each scenario, answer the following questions:

1. Does the target population change? Do the eligibility criteria still ensure that all patients could plausibly receive either treatment strategy?
2. What do the changes to censoring imply about the treatment strategy now being estimated?

3. What biases would you be concerned about introducing by this modification?
How could you address them?
4. Would these treatment strategies be feasible to implement in a clinical setting?

Scenario 1: Your colleague proposes modifying the follow-up component of the protocol to focus exclusively on treatment adherence and continuation, such that follow-up would end either if they changed treatment, experienced a study outcome (AF/Aflutter, stroke), were lost to follow-up, or the study period ended. Under the modified approach, patients would be censored when they: (1) discontinue their assigned GLP-1RA or SGLT2 inhibitor, (2) switch to the comparison medication, (3) add the comparison medication during follow-up, or (4) discontinue levothyroxine. They argue this approach maintains focus on the true per-protocol effect.

1. Does the target population change? Do the eligibility criteria still ensure that all patients could plausibly receive either treatment strategy?

Same as original study.

2. What do the changes to follow-up and censoring imply about the treatment strategy now being estimated?

We are intervening on three components: GLP-1RA, SGLT2i, and **levothyroxine**.

These strategies are static.

The effect is now a comparison of the *initiation and continuation GLP-1RA with continuation of levothyroxine* to the *initiation and continuation of SGLT2i with continuation of levothyroxine*.

3. What biases would you be concerned about introducing by this modification?
How could you address them?

Selecting individuals who are continuing on levothyroxine may be complicated in real-world data without looking into the future; investigators would want to start follow-up based on when the individual meets all eligibility criteria and has data compatible with the treatment strategy.

Including individuals with a certain amount of time left on their prescription, i.e., at baseline individuals must have ≥ 30 days remaining on their prescription, can mitigate problems that result from using future prescription information to define eligibility, treatment strategies, and start of follow-up.

4. Would these treatment strategies be feasible to implement in a clinical setting?

These strategies may not be realistic as they do not allow for individuals to change medications when clinically indicated. Someone may switch from a GLP1 to an SGLT2 if they develop an contraindication, such as pancreatitis, or indication, such as heart failure, which SGLT2i are better suited to address.

Scenario 2: Your colleague has linked the Medicare claims data to electronic health record data that includes TSH measurements. They propose redefining the eligibility criteria such that TSH is measured at baseline, and the treatment strategies to require continued levothyroxine use with TSH monitoring to adjust levothyroxine if TSH is suppressed in both treatment strategies. Patients who do not have a TSH measurement within 18 months of their most recent test would be censored, since their adherence to the monitoring strategy could no longer be verified. They argue this approach better reflects a clinically actionable strategy by incorporating guideline-recommended thyroid monitoring into the protocol definition.

1. Does the target population change? Do the eligibility criteria still ensure that all patients could plausibly receive either treatment strategy?

The target population now includes individuals who have a known TSH level, but still does not restrict based on indication/contraindications for GLP-1RA and SGLT2i.

2. What do the changes to follow-up and censoring imply about the treatment strategy now being estimated?

We are intervening on four components: GLP-1RA, SGLT2i, levothyroxine, and TSH measurement.

The strategies are now *dynamic* since levothyroxine depends on TSH levels.

The effect is now a comparison of the *initiation and continuation of GLP-1RA along with the continuation of levothyroxine based on TSH levels and measuring TSH at least every 18 months* to the *initiation and continuation of SGLT2i along with the continuation of levothyroxine based on TSH levels and measuring TSH at least every 18 months*.

3. What biases would you be concerned about introducing by this modification? How could you address them?

Selecting individuals at baseline based on future TSH measurements could introduce immortal time and selection bias.

We would want to include all individuals who meet the eligibility criteria regardless of whether they have data on TSH measured every 18 months.

4. Would these treatment strategies be feasible to implement in a clinical setting?

These strategies may be more realistic than Scenario 1, since they allow changes to levothyroxine based on TSH, which is compatible with clinical practice guidelines, however, no changes are allowed for GLP1 or SGLT2.

Scenario 3: Your colleague suggests the SGLT2 inhibitors is less relevant and wants to focus on GLP-1RA alone. They propose redefining the eligibility criteria such that the requirement for ≥ 6 months of levothyroxine is removed, and the treatment strategies to compare initiation of GLP-1RA within 3 months of the first levothyroxine prescription versus initiation of GLP-1RA within 6 months of the first levothyroxine prescription. They argue this approach addresses a more clinically relevant question about the optimal timing of GLP-1RA initiation relative to hypothyroidism treatment.

Hint: Would the treatment strategies be indistinguishable when eligibility criteria are met?

1. Does the target population change? Do the eligibility criteria still ensure that all patients could plausibly receive either treatment strategy?

The target population now includes individuals initiating levothyroxine without a contraindication for GLP-1RA.

Individuals are eligible to receive either treatment strategy, noting, that since we did not make restrictions based on A1c, individual with higher glucose levels at baseline may be more likely to start GLP-1RA earlier.

2. What do the changes to follow-up and censoring imply about the treatment strategy now being estimated?

We are intervening on four components: GLP-1RA and levothyroxine.

The strategies are now *dynamic* since GLP-1RA use depends on time and development of pancreatitis and medullary thyroid cancer.

The effect is now a comparison of the *addition of GLP-1RA within 3 months of levothyroxine initiation vs addition of GLP-1RA within 6 months of levothyroxine initiation and under both strategies, continuation on both until development of pancreatitis or medullary thyroid cancer.*

3. What biases would you be concerned about introducing by this modification?
How could you address them?

Selecting individuals at baseline based on future GLP-1RA initiation could introduce immortal time and selection bias.

We would want to include all individuals who initiate levothyroxine and meet eligibility criteria regardless of GLP-1RA initiation post-baseline.

4. Would these treatment strategies be feasible to implement in a clinical setting?

These strategies may be more realistic than Scenario 1, since they allow changes to levothyroxine based on TSH, which is compatible with clinical practice guidelines, however, no changes are allowed for GLP1 or SGLT2.

Summary

In this Exercise, we learned how to map elements of a target trial to observed data, and understand implications of changing different protocol components.